Confidence Intervals and One-Sample Tests

October 3, 2022

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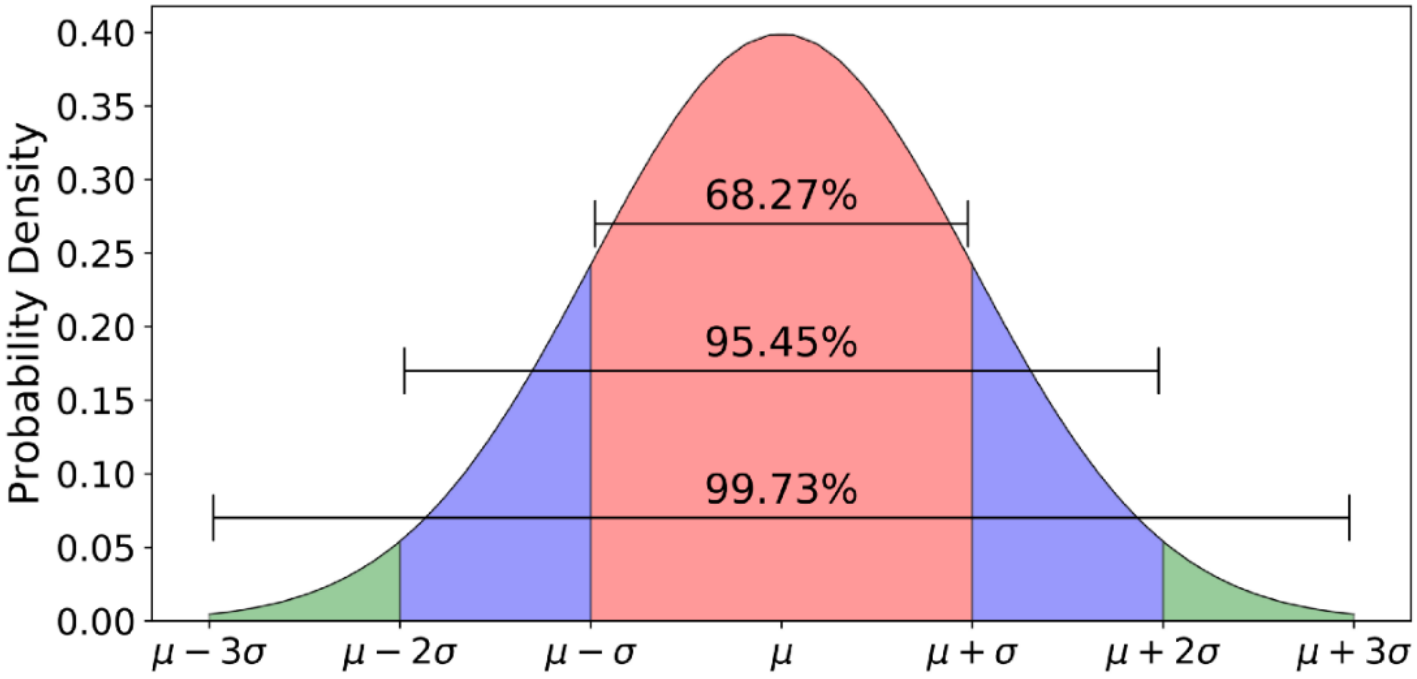
2. The sampling distribution would be normally distributed around the mean
3. The standard deviation of the sampling distribution is $\sigma_{\bar{Y}} = \frac{s_Y}{\sqrt{n}}$
4. And this would be true *even if the distributions of each of the samples is not normal*

The Normal Distribution



$\mu =$ Mean

$\sigma =$ standard deviation

68-95-99 Rule: 68% of a normal distribution falls between $\mu - \sigma$ and $\mu + \sigma$, 95% falls within $\mu - 2\sigma$ and $\mu + 2\sigma$, etc.

Some Important Reminders

- We can examine the distribution of a variable and then calculate descriptive statistics to describe that distribution (e.g., mean, standard deviation)
- But variables (usually) represent data from a sample, not a population.
- We can also examine a *sampling distribution*, which is not a distribution of the *values* of a variable, but rather...

Some Important Reminders

- We can examine the distribution of a variable and then calculate descriptive statistics to describe that distribution (e.g., mean, standard deviation)
- But variables (usually) represent data from a sample, not a population.
- We can also examine a *sampling distribution*, which is not a distribution of the *values* of a variable, but rather... a distribution of *statistics* from multiple samples.
- Finally, remember that probability distributions can look basically the same as frequency distributions, but with the probability on the y-axis instead of frequency.

Problem Example: Dice Rolling (Again)

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$$\bar{Y} = \frac{1+2+3+4+5+6}{6} = \mathbf{3.5}$$

No one actually had this... but if we did many samples and collected all the means, that distribution would be normal.

Standard Error (of the Mean)

In a sampling distribution, the standard error of the mean is the standard deviation of a sample divided by the square root of the sample *size* (not the number of samples)

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So, if standard deviation of sample is (for example), 2, then

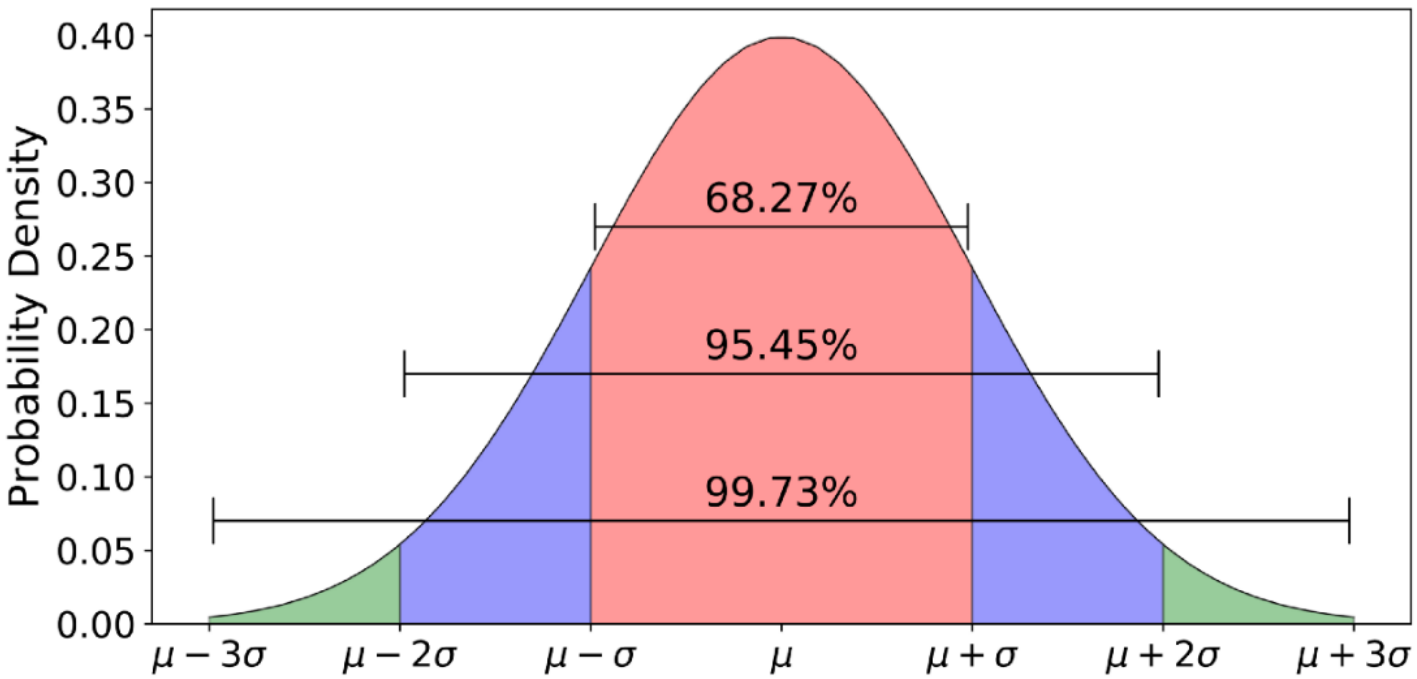
$$\sigma_{\bar{Y}} = \frac{2}{\sqrt{15}} = 0.52$$

Z-Scores

In a normal distribution, a z-score represents the number of standard deviations an observation falls from the mean.

$$z = \frac{Observation - Mean}{StandardDeviation}$$

The Normal Distribution



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Confidence Intervals

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A **Confidence Interval** for a point estimate is an interval of numbers within which the parameter (true population value) is believed to fall. The probability that this method produces an interval that contains the parameter is called the *confidence level*.

1. Choose a degree of confidence
2. Find corresponding z-score
3. Find **margin of error**: Multiply z-score and $\sigma_{\bar{Y}}$ (Standard Error)
4. Add <u>and</u> subtract that number from sample mean.

Interpreting Confidence Intervals

Correct interpretation of a 95% CI: If multiple samples were drawn from the same population and a 95% CI calculated for each sample, we would expect the population parameter (e.g., mean) to be found within 95% of these CIs.

Less correct but often used interpretation: We can be 95% confident that the population parameter (e.g., mean) is within those CIs.

Example

In 2017, survey conducted for President Trump's approval. 900 randomly selected adults, 43% approved, 52% disapproved, 5% were unsure. What would be a good estimate for the population approval rating?

1. What is the sample "mean" for approval, if we assume $approve = 1$ and everyone else is 0?

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1. What is the sample "mean" for approval, if we assume *approve* = 1 and everyone else is 0?

$$\bar{Y} = \frac{\sum(387x1) + (513x0)}{900} = 0.43$$

2. What is the sample standard deviation?

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2. What is the sample standard deviation? 0.49
3. What is the standard error of the mean (measure of uncertainty about mean)?

$$\sigma_{\bar{Y}} = \frac{0.49}{\sqrt{900}} = 0.016$$

4. If we want to be 95% confident:

$$\bar{Y} + (2\sigma_{\bar{Y}}) = 0.43 + (2x0.016) = 0.462$$

$$\bar{Y} - (2\sigma_{\bar{Y}}) = 0.43 - (2x0.016) = 0.398$$

Interpretation

(Generally) based on our sample mean estimate, we are 95% confident that the true population approval rating is between 39.8% and 46.2%

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Can we be confident that the population mean is less than 51%? This is what a hypothesis test is...

Hypothesis Testing

Use statistics to evaluate whether a population parameter differs from some value

Null Hypothesis: Assumption that a value is not different from some hypothetical value

Alternative Hypothesis: Claim that a value is different from some hypothetical value

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When we test hypotheses, we assume the null, and then try to reject it using our data.

Two Types of One-sample t-test

Two-sided (non-directional): Evaluate whether parameter is different from hypothetical value in either direction

- Was President Trump's true approval rating 20%? 70%? 45%?

One-sided (directional): Evaluate whether parameter is either greater than or less than a hypothetical value

- Was President Trump's true approval rating greater than 50%?

Imperfections of Samples

Problem: our estimates of the standard error and mean are still estimates from a sample, so it too won't be perfect.

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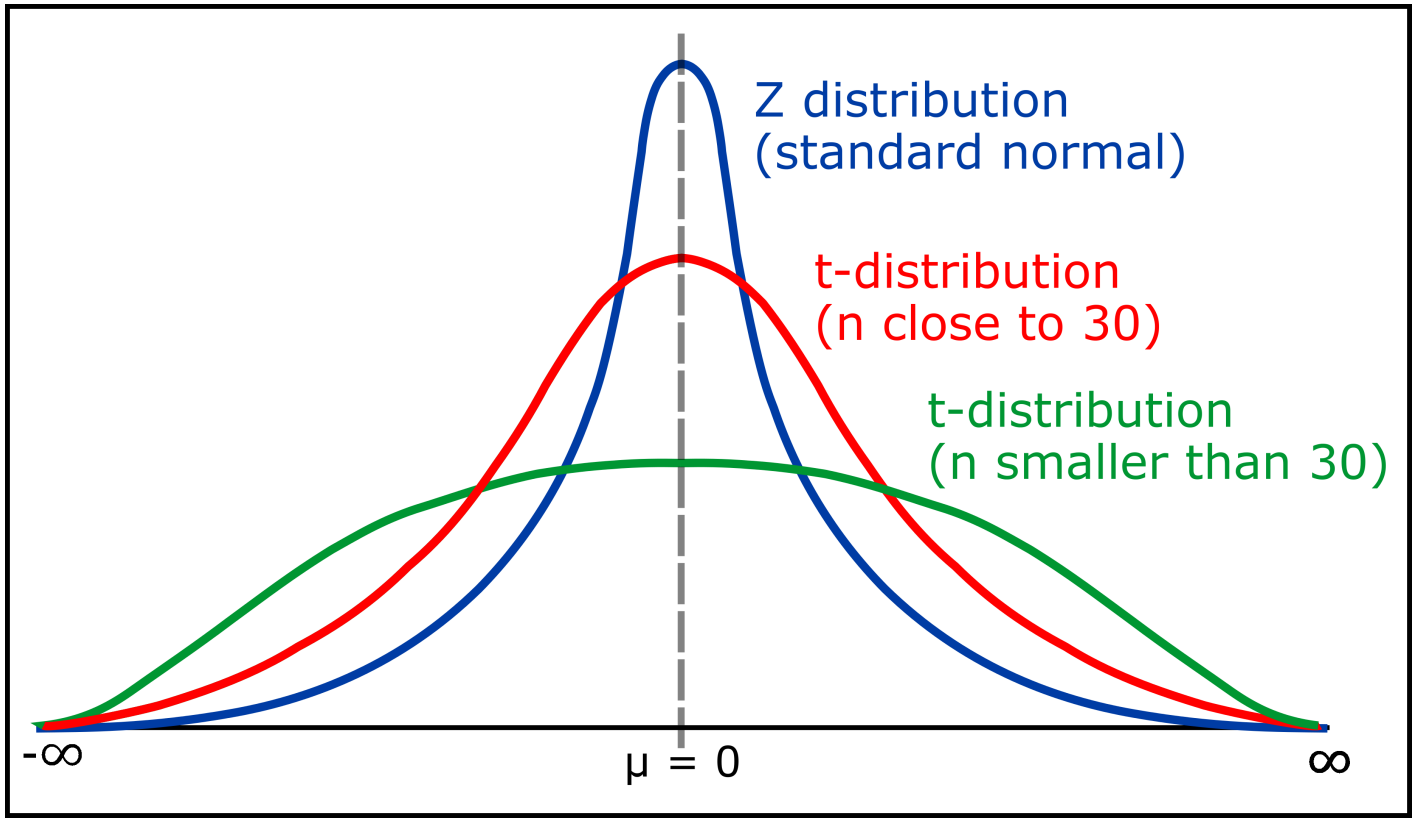
What happens to our confidence intervals if our sample sizes are large/small?

When n is small, our estimated standard errors might be way off. To account for this extra potential error, we often use a *t-score* rather than a *z-score* to test hypotheses

The t-distribution



Very similar to normal, but standard deviation depends on *degrees of freedom*: $n - 1$

The t-distribution

TABLE 5.1: Part of Table B Displaying *t*-Scores. The scores have right-tail probabilities of 0.100, 0.050, 0.025, 0.010, 0.005, and 0.001.

	Confidence Level					
	80%	90%	95%	98%	99%	99.8%
df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	$t_{.001}$
1	3.078	6.314	12.706	31.821	63.657	318.3
10	1.372	1.812	2.228	2.764	3.169	4.144
28	1.313	1.701	2.048	2.467	2.763	3.408
30	1.310	1.697	2.042	2.457	2.750	3.385
100	1.290	1.660	1.984	2.364	2.626	3.174
infinity	1.282	1.645	1.960	2.326	2.576	3.090

One-sample t-test

Luckily, t-scores function similarly to z-scores!

Calculating Confidence Intervals with t-statistics follows the same procedure:

$$\bar{y} \pm t_{\alpha}(\sigma_{\bar{Y}})$$

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Calculating Confidence Intervals with t-statistics follows the same procedure:

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Using these confidence intervals, we can evaluate whether the hypothetical value also falls within them

- In other words, *we cannot rule out that the hypothetical value is different from the observed one*

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5. So my test statistic is $t = 2$.

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3. The alternative hypothesis is that the mean is not 0.
4. I want to be confident at the 95% level,
5. So my test statistic is $t = 2$.
6. Since 0 falls outside my calculated confidence intervals, I can reject the null hypothesis that the true mean is different from 0 at the 95% level

Formulating a Hypothesis Test

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2. The null hypothesis is that the mean is 0.
3. The alternative hypothesis is that the mean is not 0.
4. I want to be confident at the 95% level,
5. So my test statistic is $t = 2$.
6. Since 0 falls outside my calculated confidence intervals, I can reject the null hypothesis that the true mean is different from 0 at the 95% level
   - OR, since 0 falls within my calculated confidence intervals, I cannot reject the null hypothesis that the true mean is different from 0

Just a little R today...

Import the ANES data by running the following code:

anes< −read.csv(`https://raw.githubusercontent.com/Brandon-Bolte/PLSC309_F22/main/ANES.csv`)

Next Class

Combine Statistical Significance/Difference of Means and Chi-square Test days

- Read: KW pages 163-178
- Skim: Licklider reading